

REPEATED SIGNIFICANCE TESTS FOR A LOCATION PARAMETER OF AN EXPONENTIAL DISTRIBUTION

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SUMMARY

Repeated significance tests for a location parameter of an exponential distribution are studied asymptotically. Approximations are given for the significance level, expected sample size and variance of the sample size. Asymptotic expansions up to terms tending to zero are obtained for the expected value and variance of the sample size. A comparison is made between the proposed sequential test and a fixed-sample size likelihood ratio test. Some Monte Carlo results are also given.

1. Introduction

Let P_θ denote a probability measure under which the random variables $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) with density function

$$f_\theta(x) = \exp \{-(x-\theta)\} I(x \geq \theta)$$

where $I(A)$ denotes the indicator function of the event A . We wish to decide between $H_0: \theta = 0$ and $H_1: \theta > 0$ on the basis of a sequential procedure with prescribed maximum sample size.

Let $M_n = \min(X_1, X_2, \dots, X_n)$, and define for a given function $f(b)$ and a positive integer $m = m(b)$ to be specified, the following stopping rule:

$$(1.1) \quad N = N_b = \inf \{n \geq 1 : (n-1)M_n > f(b)\}$$

$$= \infty \text{ if no such } n \text{ occurs.}$$

The procedure is then to

$$(1.2) \quad \text{Accept } \theta > 0 \text{ at stage } n \text{ if } N = n \leq m$$

$$\text{Accept } \theta = 0 \text{ at stage } m \text{ if } N > m,$$

and its sample size is $N_m = \min \{N, m\}$.

Siegmund (1977) recently considered repeated significance tests for a normal mean and presented an asymptotic analysis of his test procedure. Our first result concerns the asymptotic behaviour of the sequential procedure defined above when $\theta = 0$. More specifically, we obtain the following theorem:

Theorem 1:

If $0 < f(b) \rightarrow \infty$ as $b \rightarrow \infty$, then for each real number $t > 0$, we have as $b \rightarrow \infty$

$$(1.3) \quad P_0(\text{Reject } H_0) \sim f(b) \exp \{-f(b)\} \int_{1/t}^{\infty} x^{-1} \exp(-x) dx = \alpha \text{ (say)}$$

$$(1.4) \quad E_0(N_m) = m - f^2(b) \exp \{-f(b)\} \int_{1/t}^{\infty} x^{-1} (t - x^{-1}) \exp(-x) dx + o(f^2(b) \exp(-f(b)))$$

where

$$(1.5) \quad m = [f(b)t] \text{ and } [a] \text{ is the largest integer less than or equal to } a.$$

Our second result contains an asymptotic expansion for $E_{\theta}(N_m)$ and $\text{Var}_{\theta}(N_m)$, when $\theta > 0$, up to terms which vanish as $b \rightarrow \infty$. Similar results were recently obtained by Woodroffe (1977), Lai and Siegmund (1979) and Chow, Hsiung and Lai (1979), for stopping times based on sample sums.

Theorem 2:

If $0 < f(b) \rightarrow \infty$ as $b \rightarrow \infty$, then for each $\theta > 0$ we have as $b \rightarrow \infty$

$$(1.6) \quad E_{\theta}(N) = \theta^{-1} f(b) + 1 - \theta^{-1} + o(1).$$

$$(1.7) \quad \text{Var}_{\theta}(N) = \theta^{-2} + o(1).$$

Moreover, if $\theta > 1/t$, then as $b \rightarrow \infty$

$$(1.8) \quad E_{\theta}(N_m) = \theta^{-1} f(b) + 1 - \theta^{-1} + o(1).$$

$$(1.9) \quad \text{Var}_{\theta}(N_m) = \theta^{-2} + o(1).$$

The proofs of Theorem 1 and Theorem 2 will be given in Section 2 and Section 3 respectively. In Section 4 a comparison is made between the proposed sequential test and a fixed-sample size likelihood ratio test. Section 5 contains some Monte Carlo results.

2. Proof of Theorem 1

Define, for each $b > 0$, the functions

$$(2.1) \quad g(n) = n(M_n - \theta), h_b(n) = -\theta n + f(b) + \theta, n = 1, 2, \dots$$

The following lemma will be used in the proof of Theorem 1:

Lemma 2.1:

For each $\theta > 0$,

$$(2.2) \quad N \leq 2 + \theta^{-1} f(b) \text{ a.s.}(P_\theta)$$

$$(2.3) \quad g(N) \geq h_b(N) \text{ a.s.}(P_\theta).$$

$$(2.4) \quad N\theta/f(b) \rightarrow 1 \text{ a.s.}(P_\theta) \text{ as } b \rightarrow \infty, E_\theta(N) \sim \theta^{-1} f(b) \text{ as } b \rightarrow \infty. \text{ Hence the set } \{N/f(b)\}_{b>0} \text{ is uniformly integrable.}$$

$$(2.5) \quad g(N) \xrightarrow{D} X \text{ under } (P_\theta) \text{ as } b \rightarrow \infty, \text{ where } X \text{ is a random variable with density function } \exp(-x) \text{ for } x > 0. (\xrightarrow{D} \text{ denotes convergence in distribution}).$$

$$(2.6) \quad h_b(N) \xrightarrow{D} X \text{ under } (P_\theta) \text{ as } b \rightarrow \infty.$$

$$(2.7) \quad (g(N), h_b(N)) \xrightarrow{D} (X, X) \text{ under } (P_\theta) \text{ as } b \rightarrow \infty.$$

Proof:

$$(2.2) \text{ Since } M_n \geq \theta \text{ a.s. } (P_\theta), \text{ the proof follows immediately from the definition of } N.$$

$$\begin{aligned} (2.3) \quad g(N) - h_b(N) &= NM_N - f(b) - \theta \\ &= (N-1)M_N - f(b) + M_N - \theta \\ &> 0 \end{aligned}$$

by using the definition of N .

$$(2.4) \text{ The proof follows immediately from Lemma 1 and Lemma 2 of Chow and Robbins (1965).}$$

(2.5) This is a direct consequence of (2.4), the uniform continuity in probability of the sequence $\{n(M_n - \theta)\}$ and Theorem 1 of Anscombe (1952).

(2.6) This is a direct consequence of (2.4), (2.5) and a Theorem of Swanepoel and Geertsema (1976a).

(2.7) First, consider the case where $0 < x \leq y$. Therefore, using (2.3) and (2.5) we get

$$\begin{aligned}
 (2.7.1) \quad & P_\theta(g(N) \leq x, h_b(N) \leq y) \\
 &= P_\theta(g(N) \leq x) \\
 &\rightarrow P_\theta(X \leq x).
 \end{aligned}$$

Now, consider the case where $x > y > 0$. Using (2.6), it follows that

$$\begin{aligned}
 (2.7.2) \quad & \limsup_{b \rightarrow \infty} P_\theta(g(N) \leq x, h_b(N) \leq y) \\
 &\leq \limsup_{b \rightarrow \infty} P_\theta(h_b(N) \leq y) \\
 &= P_\theta(X \leq y).
 \end{aligned}$$

Also, by (2.3) and (2.5) we obtain

$$\begin{aligned}
 (2.7.3) \quad & \liminf_{b \rightarrow \infty} P_\theta(g(N) \leq x, h_b(N) \leq y) \\
 &\geq \liminf_{b \rightarrow \infty} P_\theta(g(N) \leq y, h_b(N) \leq y) \\
 &= \liminf_{b \rightarrow \infty} P_\theta(g(N) \leq y) \\
 &= P_\theta(X \leq y)
 \end{aligned}$$

The proof of part (2.7) of the Lemma now follows from (2.7.1), (2.7.2) and (2.7.3).

Proof of Theorem 1:

The following proof makes use of ideas similar to those of Lai and Siegmund (1977).

Define a probability measure Q by

$$(2.8) \quad Q(A) = \int_0^\infty P_\theta(A) \exp(-\theta) d\theta$$

If $Q^{(n)}(P_o^{(n)})$ denotes the restriction of $Q(P_o)$ to the space of $X_1, X_2, \dots, X_n$, then it is easy to verify that

$$(2.9) \quad dQ^{(n)} = L_n dP_o^{(n)}$$

where

$$(2.10) \quad L_n = \{ \exp((n-1)M_n) - 1 \} / (n-1).$$

From (2.8) and (2.10) we obtain

$$(2.11) \quad \begin{aligned} P_o(N \leq f(b)t) \\ = \int_o^\infty \{ N \leq f(b)t \}^{(N-1)/ \{ \exp((N-1)M_N) - 1 \}} dP_\theta \} \exp(-\theta) d\theta \\ = f(b) \exp \{ -f(b) \} \int_o^\infty [\int_\Omega J_b(N) dP_\theta] \exp(-\theta) d\theta, \text{ where} \end{aligned}$$

$$(2.12) \quad J_b(N) = I(N \leq f(b)t)(N-1)(f(b))^{-1} \exp(f(b)) / \{ \exp(N-1)M_N - 1 \}$$

and Ω denotes the sample space. By using the definition of N it follows that for all sufficiently large b

$$(2.13) \quad 0 \leq J_b(N) \leq 2tI(N \leq f(b)t) \leq 2t.$$

Hence, by (2.4) of Lemma 2.1, (2.13) and the dominated convergence theorem we get for each $\theta < 1/t$

$$(2.14) \quad \lim_{b \rightarrow \infty} \int_\Omega J_b(N) dP_\theta = 0$$

Also, by using (2.4) and (2.7) of Lemma 2.1, it follows that $J_b(N) \rightarrow \theta^{-1}$ in P_θ -measure as $b \rightarrow \infty$ for each $\theta > 1/t$. Hence, by (2.13) and the dominated convergence theorem, for each θ with $\theta > 1/t$

$$(2.15) \quad \lim_{b \rightarrow \infty} \int_\Omega J_b(N) dP_\theta = \theta^{-1}$$

Moreover, $\int_\Omega J_b(N) dP_\theta \leq 2t$ for all θ by (2.13). Hence the proof of the first part of the theorem follows from (2.11), (2.14), (2.15) and the dominated convergence theorem. The proof of the second part of the theorem follows in a similar way and the details are omitted.

3. Proof of Theorem 2

We first need the following lemma:

Lemma 3.1:

The set $\{ | - \theta N + f(b) + \theta |^p \}_{b>0}$ is uniformly integrable for all $\theta > 0$ and $p > 0$.

Proof

According to a result of Bickel and Yahav (1968) it is sufficient to prove that

$$(3.1) \quad \sum_{m=m_0}^{\infty} \sup_{b>0} P_{\theta}(|h_b(N)|^p > m) < \infty$$

for m_0 sufficiently large. Now, using (2.2) of Lemma 2.1 we get for all $b > 0$

$$(3.2) \quad P_{\theta}(N > \theta^{-1}f(b) + 1 + \theta^{-1}m^{1/p}) \approx 0 \text{ for all sufficiently large } m. \text{ Furthermore,}$$

$$(3.3) \quad \begin{aligned} & \sup_{b>0} P_{\theta}(N < \theta^{-1}f(b) + 1 - \theta^{-1}m^{1/p}) \\ & \leq \sup_{b>0} P_{\theta}(N \leq x(b)) \\ & = \sup_{\{b: m \leq f(b)^p\}} P_{\theta}(N \leq x(b)) \end{aligned}$$

where $x(b) = [\theta^{-1}f(b) + 1 - \theta^{-1}m^{1/p}]$. Also, if $y(b) = [\delta x(b)]$, for some constant $0 < \delta < 1$, we obtain for all b such that $m \leq f(b)^p$

$$(3.4) \quad \begin{aligned} & P_{\theta}(y(b) < N \leq x(b)) \\ & \leq P_{\theta}((n-1)M_n > f(b) \text{ for some } y(b) < n \leq x(b)) \\ & \leq P_{\theta}(x(b)M_{y(b)+1} > f(b)) \\ & = \exp \{ -(f(b)/x(b) - \theta)(y(b)+1) \} \\ & \leq \exp \{ \delta(\theta+1) - \delta m^{1/p} \} \end{aligned}$$

for all sufficiently large b and m . Define

$$(3.5) \quad S_n = \{ \exp((n-1)(M_n - \theta)) - 1 \} / (n-1), n = 2, 3, \dots$$

According to Theorem 3.1 of Swanepoel and Geertsema (1976b) $\{S_n, P_\theta, n \geq 2\}$ is a martingale sequence with expectation one. Using a well-known martingale inequality we get for b sufficiently large and satisfying the inequality $m \leq f(b)^P$

$$\begin{aligned}
 (3.6) \quad & P_\theta(N \leq y(b)) \\
 &= P_\theta((n-1)M_n > f(b) \text{ for some } 1 \leq n \leq y(b)) \\
 &\leq (y(b)-1) / \{ \exp(f(b)-\theta(y(b)-1))-1 \} \\
 &\leq \delta \theta^{-1} \exp(-m^{1/P}(1+\delta)/2).
 \end{aligned}$$

Hence (3.1) follows from (3.2), (3.3), (3.4) and (3.6).

Proof of Theorem 2:

The proof of (1.6) now follows from (2.6) of Lemma 2.1 and Lemma 3.1. Putting $p = 2$ in Lemma 3.1 and using (2.6) of Lemma 2.1 once again together with (1.6) we obtain the asymptotic formula (1.7) for $\text{Var}_\theta(N)$ immediately after using the relation $E_\theta \{h_b(N) - E_\theta h_b(N)\}^2 = \theta^2 \text{Var}_\theta(N)$. Note that (2.2) of Lemma 2.1 implies that $N_m = N$ a.s. for all sufficiently large b and for all $\theta > 1/t$. Hence, (1.8) and (1.9) of Theorem 2 follow directly from (1.6) and (1.7).

Remarks:

(3.7) The methods of this article can be used to obtain results similar to those in Theorems 1 and 2 for stopping variables such as that defined in (1.1) with $f(b)$ replaced by $f(b) + c_n$ for certain sequences c_n , such as $c_n = \log n$.

(3.8) The stopping rule considered by Swanepoel and Geertsema (1976b) can be studied using the same methods as in Theorems 1 and 2. Certain properties of the stopping rule described there by means of Monte Carlo studies and otherwise can thus be established rigorously.

4. Comparison between proposed sequential test and a fixed-sample size likelihood ratio test

The likelihood ratio test (referred to as procedure F) for testing $H_0: \theta=0$ against $H_1: \theta > 0$ with significance level α is given by

$$\begin{aligned}
 (4.1) \quad & \text{Reject } H_0 \text{ if } M_n \geq (\log \alpha^{-1})/n \\
 & \text{Accept } H_0 \text{ if } M_n < (\log \alpha^{-1})/n
 \end{aligned}$$

The OC-function is given for each $\theta > 0$ and sample size n_0 by

$$(4.2) \quad OC_F(\theta) = 1 - \alpha \exp(n_0 \theta) \text{ if } \theta < (\log \alpha^{-1})/n_0 \text{ and } OC_F(\theta) = 0 \text{ otherwise.}$$

Define a risk function of procedure F by

$$(4.3) \quad R(\theta, F) = OC_F(\theta) + cn_0$$

where the sampling cost is c units per observation.

Also, define a risk function of the sequential procedure (referred to as procedure S) by

$$(4.4) \quad R(\theta, S) = OC_S(\theta) + cE_\theta \{ \min(N, n_0) \}$$

This means we use $m = n_0$ as truncated point for procedure S. Estimated values of $OC_S(\theta)$ and $E_\theta \{ \min(N, n_0) \}$ are given in Table 3 for various values of θ when $n_0 = 50$. Let for each $\theta > 0$ and $c > 0$

$$(4.5) \quad R(\theta, c) = R(\theta, F) - R(\theta, S).$$

For each $\theta > 0$, let $c_0 \equiv c_0(\theta)$ be the solution of $R(\theta, c) = 0$. Since $R(\theta, c)$ is increasing in c , it follows that $R(\theta, c) \geq 0$ for all $c \geq c_0$. As an example, we give the values of c_0 when $n_0 = 50$ and $\alpha = 5\%$ in Table 1 below.

Table 1

θ	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09	0,10
$c_0(\theta)$	0,0088	0,0226	0,0510	0,0777	0,1183	0,1700	0,1201	0,0731	0,0317	0,0000

5. Mont Carlo studies

An important question which now arises is: how well do the asymptotic results in Theorems 1 and 2 describe the behaviour of the truncated sequential test? This question was investigated by means of limited Monte Carlo Studies. Each Monte Carlo result was based on 10 000 repetitions. Define, for $b > 0$

$$(5.1) \quad f(b) = b + \log b$$

so that (1.3) and (1.4) of Theorem 1 then become

$$(5.2) \quad P_0(\text{Reject } H_0) \sim \exp(-b) \int_{1/t}^{\infty} x^{-1} \exp(-x) dx \text{ as } b \rightarrow \infty$$

$$(5.3) \quad E_0(N_m) = m - \exp(-b) \int_{1/t}^{\infty} x^{-1} (t-x)^{-1} \exp(-x) dx + o(\exp(-b)).$$

Recall that

$$(5.4) \quad m = [(b + \log b)t]$$

Set the right-hand side of (5.2) equal to $\alpha = 5\%$ and solve for b in terms of t . It then follows from (5.4) that $m = 50$ for $t = 10,4$. The Monte Carlo experiments were based on these values of b , t and m . We found the Monte Carlo estimate of $P_0(\text{Reject } H_0)$ to be $0,0510 \pm 0,0022$ (the $\pm$ value is one standard error). Also, when $m = 100$ and $t = 19,49$ the Monte Carlo estimate of $P_0(\text{Reject } H_0)$ was found to be $0,0505 \pm 0,0022$. The Monte Carlo experiments therefore seem to indicate that the actual probability of type I error is quite close to the asymptotic probability of type I error.

Furthermore, if $m = 50$ and $t = 10,40$ the right-hand side of (5.3) (ignoring the $o(\exp(-b))$ term) is equal to 48,8, while the Monte Carlo estimate of $E_0(N_m)$ was found to be $48,50 \pm 0,07$. Also, when $m = 100$ and $t = 19,49$ the right-hand side of (5.3) is equal to 97,2 while the Monte Carlo estimate of $E_0(N_m)$ was found to be $96,7 \pm 0,15$. Once again, we conclude that the approximation provided by (5.3) appears quite accurate.

Table 2

θ	Asymptotic theory from (1.8)	Monte Carlo
0,12	33,2	32,1 $\pm$ 0,09
0,14	28,6	27,9 $\pm$ 0,07
0,16	25,2	24,8 $\pm$ 0,06
0,18	22,5	22,2 $\pm$ 0,06
0,20	20,3	20,3 $\pm$ 0,05
0,22	18,6	18,6 $\pm$ 0,04
0,24	17,1	17,3 $\pm$ 0,04
0,26	15,9	16,1 $\pm$ 0,04
0,28	14,8	15,1 $\pm$ 0,03
0,30	13,9	14,2 $\pm$ 0,03

Table 2 compares the right-hand side of (1.8) of Theorem 2 (ignoring the $o(1)$

term) with Monte Carlo estimates of $E_{\theta}(N_m)$ when $m = 50$ and $t = 10,40$ for various values of $\theta > 1/t$. From this it seems that the asymptotic approximation to $E_{\theta}(N_m)$, given by (1.8), is quite good.

Table 3

θ	Estimated Power			Estimated $E_{\theta}(N_m)$ -values		
0,01	0,0662	±	0,0025	48,2	±	0,08
0,02	0,0854	±	0,0028	47,8	±	0,08
0,03	0,1018	±	0,0030	47,6	±	0,08
0,04	0,1377	±	0,0034	47,0	±	0,09
0,05	0,1816	±	0,0039	46,4	±	0,10
0,06	0,2498	±	0,0043	45,6	±	0,10
0,07	0,3427	±	0,0047	44,5	±	0,11
0,08	0,4890	±	0,0050	43,0	±	0,11
0,09	0,7052	±	0,0046	40,7	±	0,11
0,10	1,0000	±	0,0000	37,9	±	0,11

Table 3 gives for each $\theta = 0,01; 0,02; \dots; 0,10$ the estimated power of the sequential procedure together with the estimated value of $E_{\theta}(N_m)$ for $m = 50$ and $t = 10,40$. These values are needed for the construction of Table 1.

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